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Sen Na

Assistant Professor, School of Industrial and Systems Engineering
Georgia Institute of Technology

Phone: (773) 580-6556 Email: senna@gatech.edu Homepage: https://senna1128.github.io

I. Earned Degrees

- **University of Chicago**, Chicago, Illinois USA — Aug 2016 - July 2021

  Ph.D. in Statistics (GPA 4.0/4.0)

  Thesis: Towards Solving Long-Horizon Nonlinear Dynamic Programs: Scalability and Robustness

  Advisors: Prof. Mihai Anitescu and Prof. Mladen Kolar

  Committee members: Prof. Lek-Heng Lim and Prof. Tengyuan Liang

- **Nanjing University**, Nanjing, Jiangsu China — Sept 2012 - June 2016

  B.S. in Mathematics (GPA 3.9/4.0, rank 1/120)

  Thesis: A Stochastic Semi-Proximal-Based Peaceman-Rachford Splitting Method

- **University of California, Davis**, Davis, California USA — Mar 2015 - Sept 2015

  Department of Statistics, exchange student (GPA 4.0/4.0)

II. Employment History

- **Georgia Institute of Technology**, Atlanta, Georgia USA — Aug 2024 - present

  H. Milton Stewart School of Industrial and Systems Engineering

  Assistant Professor

- **University of California, Berkeley**, Berkeley, California USA — Sept 2021 - July 2024

  Department of Statistics and International Computer Science Institute

  Postdoctoral Scholar

  Advisor: Prof. Michael W. Mahoney

- **Argonne National Laboratory**, Lemont, Illinois USA — June - Sept., 2018, 2019, 2020

  Mathematics and Computer Science Division

  Givens Associate in 2018, 2019, and W.J. Cody Associate in 2020

  Projects: sensitivity analysis of nonlinear dynamic programs, convergence analysis of online model predictive control, convergence analysis of offline temporal decomposition procedure, applications on power grids and energy systems, implementation of Julia/JuMP, IPOPT, etc.

III. Honors and Awards

- **MAPR Meritorious Service Award**, Mathematical Optimization Society — 2023
- **ORIE Young Researcher**, Cornell University — 2023
- **SDSS Student & Early Career Award**, ASA — 2023
- **Harper Dissertation Fellowship**, University of Chicago — 2020

  (one of the highest honors at UChicago in recognition of Ph.D. candidates' record of achievement and

professional promise.)

- **Bao-steel Scholarship**, Nanjing University 2016
  (one of the highest national honors given to undergraduate students for academic performance.)
- **Outstanding Graduate**, Nanjing University 2016
- **Exchange Program Scholarship**, University of California, Davis 2015
- **Electronics Technology Scholarship**, Nanjing University 2014
- **Aolei Scholarship**, Nanjing University 2013

IV. Research, Scholarship, and Creative Activities

A. Research Interests

- High-Dimensional Estimation & Inference
- Graphical Models & Semiparametric Models
- Large-Scale Stochastic Optimization
- Uncertainty Quantification
- Sequential Decision-Making & Optimal Control & Networks
- AI for Science: applications in biology, neuroscience, physics, and engineering

B. Refereed Publications and Submitted Articles

Abbreviations below:
COLT: Conference on Learning Theory.
ICML: International Conference on Machine Learning.
NeurIPS: Advances in Neural Information Processing Systems.

Notation:
$\star$ refers to the work completed at Georgia Tech.
† refers to Equal Contribution.
Authors displayed in red are my students at Georgia Tech when the first draft was completed.
Authors displayed in blue are my students at Other Institutions when the first draft was completed.

B1. Published and Accepted Journal Articles

[J18] $\star$ Near-Optimal Performance of Stochastic Model Predictive Control
S. Shin, **S. Na**, and M. Anitescu
To appear in Mathematics of Operations Research, 2025 Preprint

[J17] $\star$ Statistical Inference of Constrained Stochastic Optimization via Sketched Sequential Quadratic Programming
S. Na and M. W. Mahoney
Journal of Machine Learning Research, 2025 Preprint

[J16] $\star$ Panda: partially approximate newton methods for distributed minimax optimization with unbalanced dimensions
M. Xiao, C. Liu, C. Chen, J. C. S. Lui, and **S. Na**
Machine Learning, 2025 DOI: 10.1007/s10994-025-06813-1

[J15] Fully Stochastic Trust-Region Sequential Quadratic Programming for Equality-Constrained Optimization Problems
Y. Fang, **S. Na**, M. W. Mahoney, and M. Kolar

SIAM Journal on Optimization, 2024 DOI: 10.1137/22m1537862

[J14] Convergence Analysis of Accelerated Stochastic Gradient Descent Under the Growth Condition
Y.-L. Chen†, **S. Na**†, and M. Kolar
Mathematics of Operations Research, 2023 DOI: 10.1287/moor.2021.0293

[J13] A Fast Temporal Decomposition Procedure for Long-horizon Nonlinear Dynamic Programming
S. Na, M. Anitescu, and M. Kolar
Mathematics of Operations Research, 2023 DOI: 10.1287/moor.2023.1378

[J12] Inequality Constrained Stochastic Nonlinear Optimization via Active-Set Sequential Quadratic Programming
S. Na, M. Anitescu, and M. Kolar
Mathematical Programming, 2023 DOI: 10.1007/s10107-023-01935-7

[J11] Hessian averaging in stochastic Newton methods achieves superlinear convergence
S. Na, M. Dereziński, and M. W. Mahoney
Mathematical Programming, 2022 DOI: 10.1007/s10107-022-01913-5

[J10] An adaptive stochastic sequential quadratic programming with differentiable exact augmented lagrangians
S. Na, M. Anitescu, and M. Kolar
Mathematical Programming, 2022 DOI: 10.1007/s10107-022-01846-z

[J9] On the Convergence of Overlapping Schwarz Decomposition for Nonlinear Optimal Control
S. Na†, S. Shin†, M. Anitescu, and V. M. Zavala
IEEE Transactions on Automatic Control, 2022 DOI: 10.1109/tac.2022.3194087

[J8] Superconvergence of Online Optimization for Model Predictive Control
S. Na and M. Anitescu
IEEE Transactions on Automatic Control, 2022 DOI: 10.1109/tac.2022.3223323

[J7] SFGAE: a self-feature-based graph autoencoder model for miRNA–disease associations prediction
M. Ma, **S. Na**, X. Zhang, C. Chen, and J. Xu
Briefings in Bioinformatics, 2022 DOI: 10.1093/bib/bbac340

[J6] The Graph-Based Behavior-Aware Recommendation for Interactive News
M. Ma, **S. Na**, H. Wang, C. Chen, and J. Xu
Applied Intelligence, 2021 DOI: 10.1007/s10489-021-02497-x

[J5] High-dimensional index volatility models via Stein's identity
S. Na and M. Kolar
Bernoulli, 2021 DOI: 10.3150/20-bej1238

[J4] AEGCN: An Autoencoder-Constrained Graph Convolutional Network
M. Ma, **S. Na**, and H. Wang
Neurocomputing, 2021 DOI: 10.1016/j.neucom.2020.12.061

[J3] Estimating differential latent variable graphical models with applications to brain connectivity
S. Na, M. Kolar, and O. Koyejo
Biometrika, 2020 DOI: 10.1093/biomet/asaa066

[J2] Exponential Decay in the Sensitivity Analysis of Nonlinear Dynamic Programming
S. Na and M. Anitescu
SIAM Journal on Optimization, 2020 DOI: 10.1137/19m1265065

[J1] High-dimensional Varying Index Coefficient Models via Stein's Identity
S. Na, Z. Yang, Z. Wang, and M. Kolar
Journal of Machine Learning Research, 2019 Preprint

** Towards Solving Long-Horizon Nonlinear Dynamic Programs: Scalability and Robustness
S. Na
University of Chicago (PhD Thesis) (2021)

B2. Conference Presentation with Proceedings (Refereed)

[C4] ⋆ Online Covariance Estimation in Nonsmooth Stochastic Approximation
L. Jiang†, A. Roy†, K. Balasubramanian, D. Davis, D. Drusvyatskiy, and **S. Na**
Conference on Learning Theory, 2025 Preprint

[C3] Constrained Optimization via Exact Augmented Lagrangian and Randomized Iterative Sketching
I. Hong†, **S. Na**†, M. W. Mahoney, and M. Kolar
International Conference on Machine Learning, 2023 Preprint

[C2] Global Convergence of Online Optimization for Nonlinear Model Predictive Control
S. Na
Advances in Neural Information Processing Systems, 2021 Preprint

[C1] Semiparametric Nonlinear Bipartite Graph Representation Learning with Provable Guarantees
S. Na, Y. Luo, Z. Yang, Z. Wang, and M. Kolar
International Conference on Machine Learning, 2020 Preprint

B3. Other Refereed Material

[W2] ⋆ Online Statistical Inference for Proximal Stochastic Gradient Descent under Markovian Sampling
X. Du and **S. Na**
Workshop on Constrained Optimization for Machine Learning, NeurIPS, 2025

[W1] ⋆ Statistical Inference of Constrained Model Estimation via Derivative-Free Stochastic Sequential Quadratic Programming
J. Ye and **S. Na**
Workshop on Constrained Optimization for Machine Learning, NeurIPS, 2025

B4. Submitted Articles

[P9] ⋆ Online Inference of Constrained Optimization: Primal-Dual Optimality and Sequential Quadratic Programming
Y. Gao, M. K. Ng, M. W. Mahoney, and **S. Na**
arXiv preprint arXiv:2512.08948, 2025 Preprint

[P8] ⋆ Derivative-Free Sequential Quadratic Programming for Equality-Constrained Stochastic Optimization
S. Na
arXiv preprint arXiv:2510.22458, 2025 Preprint

[P7] ⋆ Overlapping Schwarz Scheme for Linear-Quadratic Programs in Continuous Time
H. Zhao, M. Anitescu, and **S. Na**
arXiv preprint arXiv:2510.04478, 2025 Preprint

[P6] ⋆ Online Statistical Inference of Constrained Stochastic Optimization via Random Scaling
X. Du, W. Zhu, W. B. Wu, and **S. Na**
arXiv preprint arXiv:2505.18327, 2025 Preprint

[P5] ⋆ High Probability Complexity Bounds of Trust-Region Stochastic Sequential Quadratic Programming with Heavy-Tailed Noise
Y. Fang, J. Lavaei, and **S. Na**
arXiv preprint arXiv:2503.19091, 2025 Preprint

[P4] ⋆ Online Covariance Matrix Estimation in Sketched Newton Methods
W. Kuang, M. Anitescu, and **S. Na**
arXiv preprint arXiv:2502.07114, 2025 Preprint

[P3] ⋆ Trust-Region Sequential Quadratic Programming for Stochastic Optimization with Random Models
Y. Fang, **S. Na**, M. W. Mahoney, and M. Kolar
arXiv preprint arXiv:2409.15734, 2024 Preprint

[P2] ⋆ Physics-Informed Neural Networks with Trust-Region Sequential Quadratic Programming
X. Cheng and **S. Na**
arXiv preprint arXiv:2409.10777, 2024 Preprint

[P1] Distributed Sequential Quadratic Programming with Overlapping Graph Decomposition and Exact Augmented Lagrangian
R. Ni, **S. Na**, S. Shin, and M. Anitescu
arXiv preprint arXiv:2402.17170, 2024 Preprint

C. Presentations

C1. Invited Seminar Presentations

[3]	UC Berkeley (biostatistics seminar)	Berkeley, Apr. 2024
[2]	Stanford University (optimization seminar)	Stanford, Mar. 2024
[1]	Cornell ORIE Young Researchers Workshop (referred)	Cornell, Oct. 2023

C2. Invited Conference Presentations

[9]	INFORMS Annual Meeting	Atlanta, Oct. 2025
[8]	International Conference on Continuous Optimization	Los Angeles, July 2025
[7]	INFORMS Annual Meeting	Phoenix, Oct. 2023
[6]	International Conference on Machine Learning	Honolulu, July 2023
[5]	SIAM Conference on Optimization	Seattle, May 2023
[4]	Symposium on Data Science & Statistics (referred)	St. Louis, May 2023
[3]	Advances in Neural Information Processing Systems	Virtual, Dec. 2021
[2]	International Conference on Machine Learning	Virtual, July 2020
[1]	Summer Student Mini-Symposium, Argonne National Laboratory	Lemont, Sept. 2018-2020

D. Grants and Contracts

D1. Pending Proposals

- **Title**: Stochastic Numerical Methods for Constrained Optimization: Computational Efficiency, Statistical Optimality, and Applications
  Agency/Company: NSF
  Total Dollar Amount: $492,339
  Role: PI
  Period of Contract: 6/1/2026 – 6/1/2029
  Share: 100%

D2. Proposals Submitted but Not Funded (Last Two years)

- **Title**: Stochastic Numerical Methods for Constrained Optimization: Computational Efficiency, Statistical Optimality, and Applications
  Agency/Company: NSF
  Total Dollar Amount: $420,115
  Role: PI
  Period of Contract: 5/1/2025 – 5/1/2025
  Share: 100%

V. Education

A. Courses Taught as Instructor

- **Georgia Tech, School of Industrial and Systems Engineering**

Semester	Number	Course Title	# Students	Rating
Spring 2025	ISyE 6663	Nonlinear Optimization	22	4.45
Fall 2025	ISyE 6414	Regression Analysis	31	4.35
Spring 2026	ISyE 7401	Advanced Statistical Modeling		
Spring 2026	ISyE 6414	Regression Analysis		

B. Courses Taught as Teaching Assistant

- **University of Chicago, Department of Statistics**

  Teaching Assistant

  - STAT376 Machine Learning and Large-Scale Data Analysis — Spring 2020
  - STAT315 Stochastic Simulation — Spring 2019
  - STAT245 Statistical Theory and Methods (II) — Winter 2021
  - STAT244 Statistical Theory and Methods (I) — Autumn 2017
  - STAT234 Statistical Models and Methods — Spring 2017, Winter 2017, Winter 2018

C. Individual Student Guidance

C1. Ph.D. Students at Georgia Tech

[3] Xinchen Du

Ph.D. in Machine Learning, Fall 2025 – Present
Research: Online Decision-Making and Generative Modeling

[2] Haoxuan Wang
Ph.D. in Machine Learning, Fall 2024 – Present
Research: Computational Statistics with Uncertainty Quantification

[1] Jianliang Ye
Ph.D. in Mathematics, Fall 2024 – Present
Research: PDE-Constrained Optimization and Control

C2. Students at Other Institutions

[4] Jingrong Guan
Undergraduate Student in Mathematics, ShanghaiTech University, Spring 2025 – Present

[3] Yufei Fan
Ph.D. in Data Sciences & Operations, University of Southern California, Fall 2025 – Present

[2] Xiaoran Cheng
Ph.D. in Statistics, Penn State University, Fall 2024 – Present

[1] Yuchen Fang
Ph.D. in Mathematics, UC Berkeley, Fall 2023 – Present

C3. M.S. Students

[4] Ilgee Hong
M.S. in Statistics, UChicago, Fall 2021 – Spring 2023
Position: Georgia Tech, ISyE PhD

[3] Simiao Jiao
M.S. in Statistics, UChicago, Fall 2021 – Spring 2023
Position: Duke, Fuqua PhD

[2] Miao Li
M.S. in Statistics, UChicago, Fall 2020 – Spring 2022
Position: Georgia Tech, ISyE PhD

[1] Heming Liu
M.S. in Statistics, UChicago, Fall 2020 – Spring 2022
Position: Northwestern, IEMS PhD

C4. Service on Thesis or Dissertation Committees

[4] Yixiao Li (GT ISyE), Dissertation Proposal — Fall 2025

[3] Zixuan Zhang (GT ISyE), Dissertation Proposal — Fall 2025

[2] Alexander Bukharin (GT ISyE), Ph.D. defense — Fall 2024

[1] Haoyun Wang (GT ISyE), Ph.D. defense — Fall 2024

VI. Service

A. Professional Contributions

A1. Referee

- *Referee Service*

  I have served as a referee for several prestigious journals in the fields of mathematics, statistics, and optimization, including:

  - Journal of Machine Learning Research
  - IMA Journal of Numerical Analysis
  - Mathematics of Operations Research
  - Mathematical Programming
  - SIAM Journal on Scientific Computing
  - SIAM Journal on Optimization
  - Journal of Computational and Graphical Statistics
  - Electronic Journal of Statistics
  - Statistics & Probability Letters
- *Reviewer* of conferences: NeurIPS, ICML, ICLR, IJCAI, AIStats etc.

B. Public and Community Service

- *Organizer* of conference sessions: SIAM Conference on Optimization, INFORMS Annual Meeting, International Conference on Continuous Optimization, INFORMS Optimization Society Conference